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Sustainability PIVOT

INDIA'S PHARMACEUTICAL SECTOR SUPPLIES MORE THAN 60% OF THE WORLD'S VACCINES AND 40% OF THE GENERIC DRUGS USED IN THE US. BUT THAT COMES AT AN ECOLOGICAL COST, PROMPTING PHARMA COMPANIES TO SHIFT TO CLEANER PRACTICES

BY NEETU CHANDRA SHARMA

AT DAWN IN HYDERABAD'S pharmaceutical belt, technician Ravi Kumar, responsible for maintaining the power utilities of a private firm, steps onto a rooftop that once echoed with the growl of diesel generators. It is a lot quieter these days—and cleaner. Rows of solar panels glint in the morning sun.

Below, wastewater generated from the manufacturing of drugs circulates through a zero-liquid discharge (ZLD) system, which ensures that no contaminated water leaves the facility. "I've seen this place go from fumes and waste to something cleaner, smarter," says 59-year-old Ravi.

This reflects a reckoning across India's pharma sector that supplies more than 60% of the world's vaccines and nearly 40% of the generic drugs consumed in the United States, where environmental responsibility is beginning to match business ambition.

Pharmaceutical manufacturing uses large amounts of water and energy. It also produces chemical waste that can harm the environment if not treated properly. In a country like India, this creates a challenge of meeting the growing demand without damaging local ecosystems or contributing to climate change.

But this growth carries a cost. The manufacture of active pharmaceutical ingredient (API)—the backbone of drug production—is one of the most resource-intensive activities. According to the global non-profit Carbon Disclosure Project's (CDP's) In-

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► **High Environmental Impact of API Manufacturing:** API production is highly energy and water intensive, flagged by the Ministry of Environment as one of the most energy-consuming industrial segments